UPSAFE: A Universal Physical Security Assessment Framework for Personal Safety Evaluation Across Multi-Hazard Environments

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Abstract*—The majority of frameworks of physical security risk assessment have focused on information technology systems, critical infrastructure or disaster population measures at aggregate levels. There is no ethically sound, calculative model of assessing the personal safety of the person affected by any hazard to physical harm, whether atmospheric pollution, natural disaster, chemical release or kinetic threat or biological outbreak, especially in the infrastructure-lacking conditions as conflict zone, disaster-impacted area or an offsite-industrial complex. In the paper, a NIST SP 800-30 aligned with ISO 31000-aligned approach has been proposed, where the individual human is the asset under protection. UPSAFE calculates a quantitative Personal Risk Score (PRS) based on an eight-step identification process, a five-by-five Likelihood, Impact Determination Matrix, and formal financial risk measures, Single Loss Expectancy (SLE) and Annualized Loss Expectancy (ALE) scaled to the physical safety domain, using adaptation to NIST quantitative risk measurement. The validation framework is based on the India Multi-City Air Quality Dataset (21 636 records, 284 cities, January 2021-March 2025), generating empirical PRS assignments, SLE and ALE estimations, and a direct comparison of the framework with the traditional CPCB Air Quality Index advisory framework of 10 major Indian cities. Findings reveal that UPSAFE rightly predicts Extreme risk in Delhi (ALE = INR 4,27,500/year) and High risk in Patna, both of which the CPCB average AQI system underestimates. The fact of PRS degradation under better PPE conditions is evidenced by sensitivity analysis.

***Keywords*—Physical Security Risk Assessment, Personal Safety, Multi-Hazard Framework, Risk Matrix, NIST SP 800-30, ISO 31000, SLE, ALE, Air Quality Risk, India AQI**

I. Introduction

Human health is at risk due to a complex combination of environmental, physical, chemical, biological and kinetic hazards. The soldier who has to work in a conflict zone and which is polluted with chemical substances, the rescue worker in the city where a flood has swept and the infrastructure is destroyed, the coal mines worker who is exposed to unhealthy levels of particulate matter, the civilian living in the city, where the Air Quality Index (AQI) consistently is above dangerous limits each of them has a personal calculus of risk to personal safety which is specific and time. The major and repeated question in all these scenarios is the same, How dangerous is it here, now, and in my case?

The available risk assessment models are mostly based on solving various questions. NIST SP 800-30 (Joint Task Force Transformation Initiative, 2012) [1] is a resource that deals with information security risk in the IT systems. The concept of enterprise risk management is also presented by ISO 31000:2018 (International Organization for Standardization, 2018) [2], but without a model of what to score in the field to use as a computer score. Security Risk Assessment Methodology [4] is aimed at safeguarding the buildings and critical infrastructure. Population-level statistic mortality is calculated by Air Pollution Health Risk Assessment frameworks [7]. Multi-hazard disaster models [8] are based at the city or geographic level. All of these frameworks do not place the individual human body at the center of the risk computation, and result in a real-time, personally actionable safety tier, with quantified financial risk metrics.

The following three convergent realities in modern hazard environments create motivation in UPSAFE. First, the National Capital Region of India registered AQI scores of over 400 (the range of which is Severe) more than 20 days over the winter of 2024, but there is no individual risk scoring system which would utilize the AQI score and provide a personal directive to a particular worker or resident commonly spending time outdoors. Second, humanitarian organisations working in areas of conflict (Ukraine 2022-present, Gaza 2023-present, Sudan 2023-present) document that the field staff there lacks a unified approach to determine their personal score on the safety framework in the face of kinetic, chemical, environmental risks. Third, the WHO (2021) [14] estimates that the annual premature death rate caused by ambient air pollution alone is 4.2 million; high-risk focus people, such as elderly and immunocompromised people and those whose work is out of doors, are not targeted on a personal scale. The acute necessity to seal this gap, with a principled, field-deployable, and computationally transparent personal risk assessment framework, drives UPSAFE. In this paper, the researchers present the following five new contributions to the aforementioned field of physical security and safety risk assessment:

- The earliest domain-neutral framework that considers the individual human as the asset to be protected and calculates a personal safety level (E/H/M/L) that is used across all eight types of physical hazards, from atmospheric pollution to kinetic threats.
- The initial risk of considering personal physical safety with the quantitative risk mathematics of NIST SP 800-30 [1].
- Single Loss Expectancy (SLE) and Annualized Loss Expectancy (ALE) method allows formal cost-benefit analysis of the personal protective interventions.
- An innovative six-dimensional vulnerability scoring tool which modulates the Likelihood rating as appropriate to individual susceptibility factors (proximity, duration, PPE, health status, awareness, infrastructure), to achieve an individual, not population based, risk.
- Empirical validation: PRS calculation, SLE calculation, ALE calculation, calculation of 10 large cities in India in reference to a real national AQI dataset (12 solid months January 2021-March 2025) (21, 636 records) [15], as well as a sensitivity analysis of PRS to protective equipment improvements.
- Comparative analysis of UPSAFE outputs to the CPCB AQI advisory system [15], by showing that UPSAFE adequately models' tail-risk exposures systematically underestimated by average-based advisories.

The rest of this paper follows the following structure. Section II provides a review of related work. Section III, featuring the eight step risk identification and assessment program. Section IV presents the Universal Threat Taxonomy described Vulnerability Dimensions. Section V present the Result and discussion. Section VI craft the conclusion and future work.

II. Related Work and Research Gap

NIST SP 800-30 Rev. 1 (Joint Task Force Transformation Initiative, 2012) [1] outlines a nine-step risk assessment procedure, which includes threat identification, vulnerability analysis, calculation of likelihood, impact, and calculation of risk severity. It proposes quantitative financial measures to prioritize organisational risk, Single Loss Expectancy (SLE) and Annualized Loss Expectancy (ALE). The framework, however, specifically aims at IT systems and information assets rather than physical surroundings or the safety of individual human beings. UPSAFE is based on NIST risk vocabulary and SLE/ALE mathematics with an information system substituted by the human body as the protected asset.

The most universal risk standard is the ISO 31000:2018 (International Organization for Standardization, 2018) [2], which explicitly addresses the following risks: safety, financial, and environmental. The ISO/IEC 31010:2019 (International Organization for Standardization, 2019) [3] offers risk assessment methods such as consequence/likelihood matrices. Nevertheless, ISO 31000 offers only principles and process guidance, without any prescription of a calculable personal risk score or a 5×5 matrix. These principles are operationalized into a particular computable model by UPSAFE.

Little (2004) [4] proposed the Security Risk Assessment Methodology that stipulates the ratings of value of assets, threat probability, and vulnerability of urban infrastructure. NERC (2025) [9] establishes critical tiering of electrical grid infrastructure. Kapusta et al. (2023) [13] tested the effectiveness of physical protection systems with the incorporation of cyber-resilience. All these frameworks are safeguards of physical structures, not of individuals.

Sahri et al. (2025) [5] used USEPA Chronic Daily Intake (CDI) and Hazard Quotient (HQ) techniques on PM 2.5 and silica exposure of workers in the ceramic industry. Mukhia et al. (2025) [6] implemented the Mobile Sense platform with HQ classification and discovered 6/10 motorcycle riders in a high-risk group. AlKakuri et al. (2025) [19] demonstrated more than 91% classification accuracy of health risk of construction workers. Floeder et al. (2024) [18] conducted a scoping review of risk assessment methods in occupational health and hygiene, confirming the Hazard Quotient as the dominant technique across 61 reviewed studies. AP-HRA was formalized by Hassan Bhat et al. (2021) [7] to influence the population-level air pollution. Kumar et al. (2025) [10] demonstrated that traditional AQi systematically underestimates multi-pollutant health risks - a vital result that UPSAFE is tackling by employing worst-case AQi and tail-risk frequency in its Likelihood rating. The methods handle single-hazard situations with epidemiological HQ indices, without an organized 5x5 personal safety table or SLE/ALE measure.

Hochrainer-Stigler et al. (2023) [8] deal with compound and casdealsng hazards-2023. Turkey-Syria earthquake causing landslides in extreme cold - generating regional aggregate risk scores for policy planning. Alhalwachi et al. (2025) [12] reviewed multi-hazard risk assessment models in the context of disaster management and smart circular economy transitions. Ward et al. (2020) [21] survey the risk assessment of natural hazards on a global basis. Ge et al. (2022) [20] modelled armed conflict risk in climate change, and it has created the environment-conflict nexus. Rajesh et al. (2025) [11] proposed a real-time ML system of air quality health risk mapping. Naz et al. (2023) [22] compared deep learning and statistical models for air pollutant prediction in urban environments, demonstrating the superior performance of deep learning on temporal data.

Bari et al. (2023) [24] summarizes AI application in disaster risk management and mentions the 2021-2024 increase in publication. Ekkirala & Ramesh (2026) [25] offers graph-based multi-hazard risk of India, with a Wayanad and Sikkim case studies - however, only regional risk scores, not individual safety levels. No review works yield any personal, actionable safety tier of a given person in real time on all types of physical hazards. The comparative landscape is summarized in Table 1.

Table 1. Comparative Analysis of Existing Frameworks vs. UPSAFE

Study / Framework	Domain	Multi-Hazard	Personal Safety	5×5 Matrix	SLE/ALE	Crisis Scenario
NIST SP 800-30 [1]	IT Systems	Yes	No	No	No	No

ISO 31000:2018 [2]	Enterprise Risk	Yes	No	No	No	No
Little [4]	Infrastructure	Yes	No	No	No	No
Sahri et al. [5]	Occupational	No	Yes	No	No	No
Mukhia et al. [6]	Outdoor Workers	No	Yes	No	No	Partial
Hassan Bhat et al. [7]	Air Pollution	No	Yes	No	No	No
Hochrainer-Stigler et al. [8]	Multi-Hazard	Yes	No	No	No	No
NERC Guidelines [9]	Power Grid	Yes	No	No	No	No
UPSAFE (Proposed Work)	**ALL Hazards**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

III. RESEARCH METHODOLOGY

After an exhaustive examination of the current risk assessment practices and limitations of such strategies in considering person-based physical safety, we present our UPSAFE framework, which expands and refines the ideas of the NIST SP 800-30 [1] and ISO 31000 [2] into a single, person-centric risk modelling architecture. Unlike traditional system-level risk models, UPSAFE model risk around individuals as the first-order asset, incorporating multidimensional vulnerability, contextual threat intelligence and dynamic environmental monitoring into an endless decision-making pipeline.

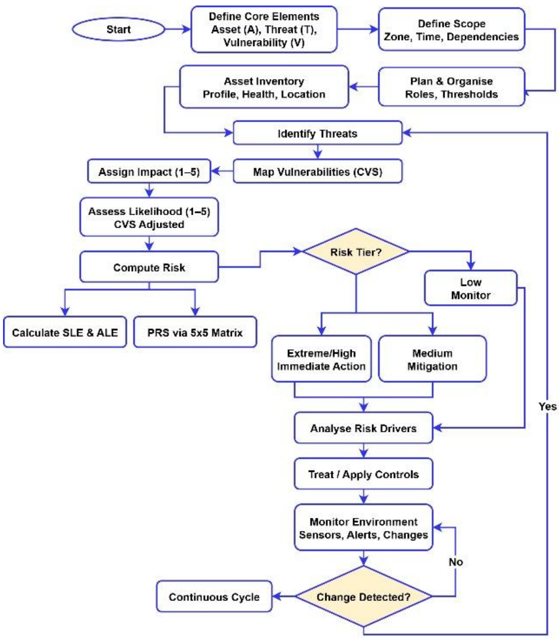

Figure 1. UPSAFE unified risk assessment and management flowgraph illustrating the integration of core risk

The framework starts by establishing fundamental components, as shown in Figure 1, whereby the Asset (A) is the person who is being assessed, and Asset Value (AV) is in monetary terms of what health-loss costs annually (derived from national health expenditure data per Pandey et al., 2021) [16]. A Threat (T) embraces any physical, chemical, biological, environmental and kinetic risk whereas Vulnerability (V) is measured by Composite Vulnerability Score (CVS), that integrates multidimensional vulnerability. These are the basic elements which determine the further formal evaluation process. The framework follows a disciplined pipeline that includes scope definition, organisational planning and asset profiling, systematic identification of active threats and mapping of respective vulnerabilities. Impact (I) and Likelihood (L) of each combination of threats and vulnerabilities is rated on a five level scale, but likelihood is determined through CVS to account for exposure-specific factor of an individual. The following parameters are then combined in a 5×5 risk matrix per NIST SP 800-30 principles [1] to calculate the Personal Risk Score (PRS) and economic parameters like Single Loss Expectancy (SLE) and Annual Loss Expectancy (ALE). The financial risk quantification approach draws on the probabilistic risk assessment methodology established in EPA Superfund guidance (Assessment, 2001) [23]. Depending on the calculated risk level, the framework recommends fitting response measures of immediate action (extreme/high risk), controlled mitigation/ongoing observation. Perhaps one of UPSAFE main contributions is the closed-loop lifecycle in which the risk analysis, the treatment, and the environmental monitoring are continuously performed. Whenever there is a perceived change in the environmental conditions or threat landscape, it is re-assessed to prevent failure in the risk management process which is adaptive and real-time.

A. *Mathematical Formulation*

1) *Composite Vulnerability Score (CVS)*

Individual vulnerability is further measured by combining six vulnerability dimensions into a scalar measure. The dimensions represent unique aspects of exposure or resilience, all presumed to be constrained on a uniform ordinal dimension. The arithmetic mean of all these six variables is termed as the Composite Vulnerability Score (CVS), and these variables equally contribute and are normalized as Eq. 1.

$$\text{CVS} = \frac{1}{6}\sum_{i=1}^{6} V_i \qquad (1)$$

Where V_1 is Proximity, V_2 is Exposure, V_3 is PPE^{-1}, V_4 is Health^{-1}, V_5 is Awareness^{-1}, and V_6 is $\text{Infrastructure}^{-1}$. To add vulnerability to probabilistic risk, the probability is conditionally modulated. The level of increased vulnerability

(CVS above 3.0) is the amplification of exposure which increases the likelihood by one unit as Eq. 2.

$$L_{\text{adjusted}} = \begin{cases} L_{\text{baseline}} + 1, & \text{if CVS} > 3.0 \\ L_{\text{baseline}}, & \text{otherwise} \end{cases} \tag{2}$$

2) Effective Operational (Atmospheric Threat T1) Function

For atmospheric threats, AQI is taken as the governing variable, and severity as a discrete mapping. The highest recorded AQI in the analysis period is used for conservative worst-case assessment [1] as Eq. 3.

$$I(\text{AQI}) = \begin{cases} 1, & \text{AQI} \le 50 \\ 2, & 50 < \text{AQI} \le 100 \\ 3, & 100 < \text{AQI} \le 200 \\ 4, & 200 < \text{AQI} \le 400 \\ 5, & \text{AQI} > 400 \end{cases} \tag{3}$$

Here, $I = \max(\text{AQI}_t), t \in [1, N]$

3) Likelihood Estimation

The likelihood is an empirical calculation based on data as an estimate of the occurrence of harmful exposure events. Pharmful counts the proportion of observations where AQI exceeds the harmful threshold Eq. 4.

$$P_{\text{harmful}} = \frac{1}{N} \sum_{i=1}^{N} I(\text{AQI}_i > 200) \tag{4}$$

This likelihood is converged to a discrete ordinal likelihood scale in order to fit within risk matrix framework as Eq. 5.

$$L_{\text{baseline}} = \begin{cases} 5, & P_{\text{harmful}} \ge 0.75 \\ 4, & 0.50 \le P_{\text{harmful}} < 0.75 \\ 3, & 0.25 \le P_{\text{harmful}} < 0.50 \\ 2, & 0.10 \le P_{\text{harmful}} < 0.25 \\ 1, & P_{\text{harmful}} < 0.10 \end{cases} \tag{5}$$

4) Personal Risk Score (PRS)

PRS is the output of the 5×5 risk mapping function, producing a categorical result aligned with ISO 31000 risk treatment principles [2] as Eq. 6.

$$\text{PRS} = R(L_{\text{adjusted}}, I), R \in \{E, H, M, L\} \tag{6}$$

5) Single Loss Expectancy (SLE)

To measure the economic impact of risk the model calculates the expected loss per incident [1] [23]. This is the Asset Value scaled by a risk-related Exposure Factor Eq. 7 and Eq. 8.

$$\text{SLE} = AV \times EF(\text{PRS}) \tag{7}$$

$$EF = \begin{cases} 0.90, & \text{PRS} = E \\ 0.65, & \text{PRS} = H \\ 0.35, & \text{PRS} = M \\ 0.10, & \text{PRS} = L \end{cases} \tag{8}$$

In this study, AV = INR 5,00,000/year, representing the per-capita annual health expenditure attributable to chronic pollution exposure in Indian urban settings, derived from national health expenditure data [15,16].

6) Annualized Loss Expectancy (ALE)

The annualized risk is calculated by adding the probability of occurrence [1, 23]. This gives an annual loss for the economy, and cost-benefit analysis of mitigation measures is possible using Eq. 9 and Eq. 10.

$$\text{ALE} = \text{SLE} \times ARO(\text{PRS}) \tag{9}$$

$$ARO = \begin{cases} 0.95, & \text{PRS} = E \\ 0.70, & \text{PRS} = H \\ 0.40, & \text{PRS} = M \\ 0.15, & \text{PRS} = L \end{cases} \tag{10}$$

IV. Threat Taxonomy and Vulnerability Dimensions

UPSAFE classifies eight mutually exclusive and exhaustive threat categories that cover the whole range of physical hazards. The taxonomy can be expanded to domain-specific additions. The complete classification, along with representative instances and primary harm pathways, is presented in Table 2.

Table 2. UPSAFE Universal Threat Taxonomy

ID	Category	Representative Instances	Primary Pathway
T1	**Environmental / Atmospheric**	Air pollution (PM2.5, PM10, NO_2, O_3), toxic gas, radiation, extreme temperature	Inhalation, dermal, thermal exposure
T2	**Natural Disaster**	Earthquake, flood, cyclone, landslide, wildfire, tsunami, GLOF	Structural collapse, drowning, displacement
T3	**Chemical / Industrial**	Gas leak, industrial fire, explosion, chemical spill, VOC emission	Toxin inhalation, burns, blast trauma
T4	**Biological / Epidemic**	Contagious disease, waterborne pathogen, vector-borne infection, bioterrorism	Infection, immune compromise, mortality
T5	**Physical / Kinetic**	Armed conflict, blast, projectile, IED, structural collapse	Trauma, mortality, psychological harm
T6	**Thermal / Climatic**	Extreme heat/cold wave, frost, humidity, WBGT index exceedance	Hyperthermia, hypothermia, cardiorespiratory stress
T7	**Radiological / Nuclear**	Radiation leak, nuclear fallout, dirty bomb, critical accident	Acute radiation syndrome, long-term cancer risk
T8	**Socio-Economic Disruption**	Supply chain failure, food/water scarcity, infrastructure breakdown	Malnutrition, dehydration, social instability

The vulnerability of each active threat is assessed in six dimensions. An increase in scores implies an increased individual susceptibility. These are summed up in the CVS to be adjusted by Likelihood adjustment. The detailed definition, measurement criteria, and scoring ranges for all six dimensions are provided in Table 3, ensuring a consistent and standardized approach to vulnerability assessment.

Table 3. UPSAFE Vulnerability Dimensions and Scoring

ID	Factor	Description / Measurement	Score Range
V1	**Proximity to Hazard Source**	Physical distance from threat origin	1–5 (1=far, 5=direct contact)
V2	**Exposure Duration**	Hours per day in hazard zone	1–5 (1≤1hr, 5 ≥8hr)
V3	**Protective Equipment**	PPE, shelter, mask, body armor	1–5 (1=full PPE, 5=none)
V4	**Individual Health Status**	Age, pre-existing conditions, immunity	1–5 (1=robust, 5=severe comorbidity)
V5	**Situational Awareness**	Knowledge of threat, evacuation plan	1–5 (1=high awareness, 5=none)
V6	**Infrastructure Availability**	Medical, emergency, communication access	1–5 (1=full access, 5=none)

The UPSAFE 5×5 Risk Determination Matrix is an expansion of traditional IT risk matrices to the physical safety area, based on NIST SP 800-30 principles [1]. Within this formulation, a category of Doomsday (Score 5) is added to represent extreme incidences (e.g. mass-casualty incidence, high levels of radiation (> 6 Gy) or direct exposure to lethal substances (e.g. nerve agents). This allows the framework to represent catastrophic underrepresented high-severity risks that are not usually represented in traditional risk assessment methodologies. Table 4 shows the entire mapping of the Likelihood and Impact, and the corresponding Personal Risk Score (PRS) tiers.

Table 4. UPSAFE 5×5 Personal Risk Determination Matrix

Likelihood \ Impact	Doomsday	Catastrophic	Moderate	Minor	Insignificant
Almost Certain	E	E	H	M	L
Likely	E	E	H	M	L
Possible	E	H	M	L	L
Unlikely	M	M	L	L	L
Rare	L	L	L	L	L

A PRS classification of Extreme (E) at the highest level of severity denotes life-threatening conditions that require an urgent intervention. This is in line with the principle of risk treatment of ISO 31000 [2], which states that any risk that is beyond the predetermined tolerance limits should be managed immediately. Interpretability of risk science outputs remains a critical challenge for field operationalisation (Thekdi & Aven, 2023) [17]. In line with this, UPSAFE recommends the deterministic response measures to be taken on each PRS tier, as presented in Table 5.

Table 5. UPSAFE PRS Tiers and Prescribed Response Actions

PRS Tier	Interpretation	Prescribed Response Action
Extreme (E)	Immediate life-threatening danger	Emergency evacuation. Cease all activity. Activate emergency medical response. Report to command immediately.
High (H)	Severe risk to health or life	Restrict movement to designated safe zones. Don maximum-rated PPE. Alert emergency services. Implement shelter-in-place protocol.
Medium (M)	Moderate risk requiring active control	Apply precautionary measures. Increase monitoring frequency. Limit daily exposure duration. Schedule medical review.
Low (L)	Acceptable risk with routine vigilance	Continue activities with standard precautions. Monitor at next scheduled interval. No immediate action required.

V. Result And Discussion

The empirical assessment of UPSAFE based on India Multi-City Air Quality Dataset [15] that consists of 21,636 bi-monthly AQI readings on 284 cities in all Indian states during January 2021 to March 2025. The dataset has more than four years of uninterrupted monitoring of air-quality in the country, and the distribution of the years is as follows: 2021 (3,566 records), 2022 (4,199 records), 2023 (5,829 records), 2024 (6,532 records), and 2025 (1,510 records up The Central Pollution Control Board (CPCB, 2020) [15] uses the computation of AQI values based on the primary pollutant concentrations, such as PM2.5, PM10, NO_2, SO_2, CO, and O_3. To provide a detailed analysis, ten representative cities were picked to experience geography and climatic differences: Delhi (Indo-Gangetic Plain), Patna (eastern plains), Lucknow (northern plains), Jaipur (northwestern arid region), Mumbai (western coast), Kolkata (eastern coast), Hyderabad (Deccan plateau), Bengaluru (southern plateau), Dehradun (Himalayan foothills; data available from April 2022), and Chennai (southern coastal region). There are between 72 and 104 observations of each city per fortnight. A standardized individual profile is used so that cross-city comparability can be obtained, which is an adult outdoor worker. The parameters of vulnerability are constant as: V_1=3.5, V_2=4.0, V_3=3.0, V_4=2.5, V_5=3.0, and V_6=2.5. Substituting these values in Eq. 1 gives a Composite

Vulnerability Score of CVS=3.083 that is over 3.0 leading to the homogenous +1 modification to the likelihood score on all cities. The AV of INR 5,00,000/year is derived from Pandey et al. (2021) [16] national health expenditure estimates for chronic pollution exposure. The results of the assessment of UPSAFE in all the chosen cities were summarized in Table 6. To obtain the Impact score (I) the highest AQI of every city and the Likelihood score (L), based on the occurrence of harmful AQI events, is then deduced and finally modified by the CVS. The resulting PRS classification is further decomposed to calculate economic risk measures such as Single Loss Expectancy (SLE) and Annualized loss expectancy (ALE) which give a qualitative and quantitative assessment of risk situation across the cities.

Table 6. UPSAFE Full Assessment Results (India AQI Dataset (Jan 2021 – Mar 2025, N = 1,040 observations)

City	N	μ AQI	Max AQI	% Poor+	Impact Level	Likelihood	I	L	PRS	SLE (INR)	ARO	ALE (INR)
Delhi	104	208.9	399.9	51.0%	Catastrophic	Likely	4	5	**E**	450,000	0.95	427,500
Patna	104	180.2	372.9	45.2%	Catastrophic	Possible	4	4	**H**	325,000	0.70	227,500
Lucknow	104	139.3	316.5	16.3%	Catastrophic	Unlikely	4	3	**M**	175,000	0.40	70,000
Jaipur	104	129.3	258.5	10.6%	Moderate	Unlikely	3	2	**L**	50,000	0.15	7,500
Mumbai	104	111.4	230.5	5.8%	Moderate	Rare	3	2	**L**	50,000	0.15	7,500
Kolkata	104	111.6	281.5	12.5%	Moderate	Unlikely	3	2	**L**	50,000	0.15	7,500
Hyderabad	104	84.8	156.9	0.0%	Minor	Rare	2	1	**L**	50,000	0.15	7,500
Bengaluru	104	74.7	113.3	0.0%	Minor	Rare	2	1	**L**	50,000	0.15	7,500
Dehradun	72	97.8	240.1	2.8%	Moderate	Rare	3	2	**L**	50,000	0.15	7,500
Chennai	104	72.6	111.8	0.0%	Minor	Rare	2	1	**L**	50,000	0.15	7,500
TOTAL/AVG	**1,040**	**130.9**	—	**14.4%**	—	—	—	—	—	—	—	**7,77,500**

Table 7 demonstrates how PRS, SLE, and ALE change when individual vulnerability parameters are modified for the Delhi baseline case [5, 6]. This analysis validates that the CVS mechanism correctly propagates individual-level protective actions into PRS changes.

Table 7. Sensitivity Analysis Delhi Baseline (PRS = E, and ALE = INR 4,27,500)

Scenario (Delhi baseline)	V_1–V_6 Sum	CVS	+1 Adj?	L_adjusted	PRS Change	SLE Change	ALE Change
V3 = 1 (Full N95 PPE)	3.5+4.0+1.0+2.5+3.0+2.5 = 16.5	2.75	No	4 → Likely	**E → H**	450,000 → 325,000	427,500 → 227,500
V3 = 3 (Base case)	3.5+4.0+3.0+2.5+3.0+2.5 = 18.5	3.083	Yes	5 → Almost Certain	**E (no change)**	450,000	427,500
V4 = 5 (Severe comorbidity)	3.5+4.0+3.0+5.0+3.0+2.5 = 21.0	3.50	Yes	5 → Almost Certain	**E (no change)**	450,000	427,500
V6 = 1 (Full infra access)	3.5+4.0+3.0+2.5+3.0+1.0 = 17.0	2.83	No	4 → Likely	**E → H**	450,000 → 325,000	427,500 → 227,500
V1=1, V3=1 (remote + PPE)	1.0+4.0+1.0+2.5+3.0+2.5 = 14.0	2.33	No	4 → Likely	**E → H**	450,000 → 325,000	427,500 → 227,500

The sensitivity analysis yields three actionable insights. First, providing N95-grade PPE (V3 = 1) reduces CVS below 3.0, eliminates the Likelihood adjustment, and downgrades PRS from E to H, saving INR 2,00,000 in ALE annually against a mask cost of approximately INR 3,600, a 56× return on investment. Second, individuals with severe comorbidities (V4 = 5) receive the same Extreme PRS but represent a higher-priority intervention target given their lower baseline resilience. Third, full infrastructure access (V6=1) also eliminates the CVS adjustment and achieves the same H-tier downgrade, confirming that improving access to emergency services is as protective as PPE provision. Table 8 presents a direct comparison between the CPCB AQI-based advisory system [15] and UPSAFE PRS for five representative cities.

Table 8. Comparative Analysis UPSAFE PRS vs. CPCB AQI Advisory System

City	CPCB Category	CPCB Advisory	UPSAFE PRS	UPSAFE Response	Key Difference
Delhi	Poor (avg 208.9)	Limit outdoors; sensitive groups	**Extreme (E)**	Immediate evacuation; mandatory PPE	UPSAFE detects Extreme AQI average understates tail risk.
Patna	Moderate (avg 180.2)	Acceptable with minor precautions	**High (H)**	Restrict movement; alert services	AQI average masks 45% days above Poor threshold.
Lucknow	Moderate (avg 139.3)	Acceptable with minor precautions	**Medium (M)**	Limit exposure; increase monitoring	Tail-risk (peak 316.5) correctly penalized.
Mumbai	Satisfactory	No advisory	**Low (L)**	Routine monitoring	Full alignment.
Chennai	Satisfactory	No advisory	**Low (L)**	Routine monitoring	Full alignment.

UPSAFE shows good operational applicability in several areas of safety significance as well as leading the state-of-the-art in the aspects of risk assessment applied in individuals. Application Field commanders are able to calculate Personal Risk Scores (PRS) of officers serving in Nuclear, Biological and Chemical (NBC) settings, where hazard information obtained by sensors map objects-to-score ratios to UPSAFE impact scores [1, 4]. Annualized Loss Expectancy (ALE) also facilitates the possibility of quantitative justification of the procurement of protective equipment. In humanitarian and disaster response settings, the framework offers a systematic methodology of determining individual safety before allowing deployment to the risky areas like floods, earthquakes, or disease outbreak [8, 12]. Likewise, in occupational health [5, 6, 18, 19], the Delhi case-study indicates that ALE can be greatly reduced with a low-cost intervention (e.g., N95 masks at INR 3,600/year), which offers a 56x payback. To implement smart city applications, UPSAFE can be coupled with an IoT-based monitoring system to produce real-time, personalized risk alerts through the synthesis of both environmental data with personal vulnerability profiles [10, 11, 22].

UPSAFE, methodologically, builds on previous research in three fundamental ways. First, in comparison with hazard quotient (HQ)-based methodologies, which focus on site-specific hazards [5, 6, 18, 19], UPSAFE offers a multi-hazard platform with the same likelihood-impact 5×5 matrix coupled with economic quantification of its possibilities through SLE and ALE. Second, unlike multi-hazard aggregation models which occur at city or regional level [8, 12, 21], UPSAFE presents a paradigm based on individuals, thus allowing risk individualization in regard to vulnerability attributes. Third, although NIST SP 800-30 [1] and ISO 31000 [2] are all about risk management in organisations, UPSAFE is ultimately explicit on the human individual as the most vital safeguarded asset whilst preserving the value of financial risks. There are three main differences detected using empirical comparison with CPCB AQI advisory data [15]. UPSAFE is more conservative in high-risk settings by placing the Extreme risk on Delhi, in contrast with the general advisory level of the CPCB. It is also more effective in hidden tail-risk exposures such as in Patna, whereby persistent surpassing of harmful AQI levels leads to High risk classification despite the moderately high average values of AQI consistent with Kumar et al., 2025 [10]. Lastly, in areas with low-risk exposure like Mumbai and Chennai, UPSAFE follows the CPCB categories, which show that the framework is not overestimating risk in safe locations.

VI. Conclusion And Future Directions

This paper introduced UPSAFE, a single-source multi-hazard multi-environment risk assessment framework that is individual-centered. The framework models the human as the key asset and calculates a Personal Risk Score (PRS) in an eight-step process within a Monitor, Identify, Analyse, and Treat lifecycle and incorporates a 5x5 likelihood-impact matrix with the SLE and economic quantification of risk based on ALE, and ISO 31000 adapted NIST SP 800-30. The validation on the India Multi-City Air Quality Dataset (2021) (21,636 records, 284 cities) indicates that UPSAFE denotes the following risk categories to Delhi (Extreme; ALE = INR 4,27,500/year), Patna (High), Lucknow (Medium), and southern/western cities (Low). The framework at the tail-risk regions captures gains lost by the mean-based advisory systems and yet is consistent in the low-risk areas. Sensitivity analysis reveals low-cost interventions (e.g., N95 masks) can significantly (analyze ALE) contribute to cost-effective decision-making. Limitations are approximate AV and EF calibration, single-threat assessment and field validation are not provided. The area of work that will be done in the future will consider real time sensor integration, wearable-based dynamic computation of CVS and the development of a Compound Personal Risk Score (CPRS) in multi-hazard surroundings. UPSAFE is designed to offer a scalable platform on which to build defence, disaster response, and occupational health and smart city systems, that can ultimately become a real-time personal safety intelligence platform.

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